SIH Problem Statements & Solutions

TID	Team Name	PS Number	Problem Statement	Proposed Solution
LRIH-01	ScrapNex	SIH26229	Kabadiwala Connect – Bringing the Informal Collector into the Formal Recycling Chain	Informal collectors aren't trapped by low-value scrap—they are trapped by a one-buyer, zero-information market that strips away their bargaining power. Our platform directly dismantles this monopsony. A Fair Price Engine eliminates price opacity by exposing real-time market ranges and trends, making lowball offers immediately obvious before any deal is struck. To shift leverage back to the collector, a competitive Recycler Bidding system replaces single-trader price setting with dynamic offers from multiple verified buyers. Built-in verification blocks unverified or suspended recyclers, protecting collectors from predatory practices while keeping toxic e-waste out of illegal processing chains. Every transaction is secured through a photo, GPS-stamped, and timestamped handover record, creating a verifiable paper trail that resolves pickup disputes and eventually establishes the transaction history needed for formal credit access. To address high-risk practices like cable burning and acid leaching, localized audio safety guidance delivers direct health awareness without relying on formal literacy. Finally, an offline-first architecture ensures the entire system functions seamlessly on a ■5,000 Android device with patchy connectivity, delivering true access where it matters most.
LRIH-02	Infintyx	SIH26105	Ai- powered continuous cyber risk quantification and investment optimization platform	We propose an AI-powered continuous cyber risk quantification and investment optimization platform that integrates security telemetry, asset criticality, and control effectiveness to continuously assess cyber risk. The platform estimates risk likelihood and financial impact, identifies major risk drivers, and provides cost-effective security investment recommendations under a defined budget. It also offers what-if simulations, risk dashboards, and explainable AI insights to help organizations prioritize cybersecurity investments and reduce potential financial losses.

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LRIH-03	CodeNova	SIH26002	Al-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)	We propose an AI-powered logistics intelligence platform designed specifically for the North Eastern Region (NER). The platform will integrate GIS maps, weather data, road accessibility information, and field reports to monitor transportation conditions in real time. It will use AI/ML to predict possible disruptions such as landslides, floods, heavy rainfall, and road blockages, estimate travel delays, and recommend safer alternative routes. The platform will also provide GPS-based vehicle tracking, automated alerts, and a centralized dashboard for monitoring essential goods and emergency routes. Multilingual support and offline data synchronization will help field officials use the system effectively in low-network areas.
LRIH-04	Mumbaivey	SIH26034	Software System to check compliance of Packaged Commodities under Legal Metrology(Packaged Commodities) Rules, 2011 by scanning products, images and labels.	Our proposal is an AI-powered software system to automate compliance checking of packaged commodities under the Legal Metrology (Packaged Commodities) Rules, 2011. Enforcement officials will upload or scan packaging, label, and product images into a user-friendly application, where OCR and AI analysis extract mandatory declarations including manufacturer details, net quantity, MRP, manufacturing/packing date, and consumer care information. Our system will validate these fields against legal requirements via a rule-based engine to identify missing, incorrect, unreadable, or non-compliant details while evaluating readability, declaration placement, and font size where feasible. It will generate detailed compliance reports with highlighted violations and image evidence, supported by a secure dashboard tracking inspection history, scanned records, compliance statistics, and violation trends for efficient enforcement management.
LRIH-05	Bharat Roots	SIH26197	Student Innovation-Ideas that showcase the rich cultural heritage and traditions of India	Bharat Heritage AI is a community-based digital heritage platform that aims to make less popular cultures of India available, accessible, and safeguardable for future generations. This platform integrates features of artificial intelligence-based cultural recognition, oral history documentation by communities, multilingual translation, geospatial interactive maps, digital heritage archive, and gaming. In the SIH prototype, we will build such a platform as an active web-based application using a regional data set, and thus show that students could learn about cultural heritage while communities could preserve their knowledge."

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LRIH-06	Debuggers	SIH26032	Farmers often face long waiting times, lack of information regarding procurement schedules, and uncertainty about procurement status.	1. Simple Slot Booking: Farmers log in with their phone number and pick a date/time to drop off crops based on available slots. 2. Smart Token System: Center officials manage incoming traffic with a single dashboard that assigns sequential token numbers to prevent overcrowding. 3. Instant SMS Alerts: Farmers receive automatic text messages on basic feature phones updating them on their appointment time and queue status. 4. Live Payment Tracker: A basic dashboard where farmers can check if their crop quality check passed and when their direct payment is credited. 5. Reduces Congestion & Waiting Time: By spreading visits across hourly time slots, the center avoids overcrowding, eliminates long vehicle lines, and keeps waiting times short.
LRIH-07	Mindpilot	SIH26207	Student Innovation-Smart education,a concept that describes learning in digital age. It enables learners to learn more effectively, efficiently, flexibly and comfortably.	We propose a smart learning website that helps students learn according to their level. It will provide study materials, AI-based doubt solving, quizzes, and progress tracking to make learning easier, flexible, and more interesting.
LRIH-08	Git Happens	SIH26169	Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication (FSOC) Terminals	We will develop - AeroAlign-AI which automates the initial coarse alignment of mobile laser communication terminals by replacing slow mechanical sweeping with computer vision and predictive AI to establish instant terminal locks. The system operates via a three-tier pipeline starting with a custom Edge-AI vision network that trains lightweight object-detection models to isolate and track distant satellites or UAVs from wide-field virtual camera feeds despite solar glare and atmospheric noise. A predictive trajectory engine then counters high-speed motion lag using a hybrid Extended Kalman Filter and LSTM recurrent network to compute and forecast target flight paths frames in advance. Finally, an automated actuation interface translates these spatial coordinates into low-latency, real-time pitch and yaw steering commands for motorized gimbals, rapidly positioning the terminal for final laser lock.

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LRIH-09	Freakuency	SIH26034	Software System to check compliance of Packaged Commodities under Legal Metrology(Packaged Commodities) Rules, 2011 by scanning products, images and labels.	We propose an AI-based software system to automate preliminary compliance checking of pre-packed commodities against the Legal Metrology (Packaged Commodities) Rules, 2011. By allowing manufacturers, consumers, and Legal Metrology officers to scan or upload product label images, our system will use OCR and image recognition to extract mandatory declarations—including product name, net weight/quantity, MRP, manufacturer/packer details, and consumer care particulars. We propose evaluating these extracted fields through a rule-based engine to flag non-compliant or missing information, displaying specific rule reference numbers, detected data, and violation reasons. We propose delivering this via two portals: an Officer portal for inspection, tracking, and report generation, and a Manufacturer/Consumer portal for uploading and checking labels. Through this automated approach, we propose to save inspection man-hours, boost awareness of statutory labeling requirements, and maintain a scalable design for future rule and vision updates.
LRIH-10	Alpha Innovators	SIH26006	Development of an Intelligent Freight Forecasting Model for Optimized Vessel Chartering and Bulk Cargo Procurement from overseas to East Coast of India	Our project develops an AI-based Freight Forecasting and Vessel Recommendation System for overseas bulk cargo procurement to India's East Coast ports. Instead of manually checking freight prices every day, our system analyzes historical freight rates, fuel prices, weather, and port constraints to predict future shipping costs. It then recommends the best booking date, the most suitable vessel type, and the optimal route to reduce logistics costs and improve supply chain efficiency. We plan to build it using Python for machine learning, SQL for data storage, and a web dashboard using HTML, CSS, JavaScript, and Flask/FastAPI or Streamlit
LRIH-11	Byte Club	SIH26204	Student Innovation-A solution/idea that can boost the current situation of the tourism industries including hotels, travel and others.	A web-based platform that aggregates unbooked hotel rooms, transit passes, and off-peak attraction tickets into dynamic, discounted bundles. It uses real-time capacity tracking and reward incentives to redirect tourist foot traffic away from overcrowded hotspots directly into underutilized local businesses.

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LRIH-12	visionX	SIH26132	Strengthening market linkages and price discovery for farmers	We propose an AI-powered digital platform that helps farmers and FPOs make better selling decisions by providing real-time mandi prices, price trends, buyer demand, quality requirements, transport and storage information. The system will use market and historical data to recommend the best market, buyer and suitable time to sell based on expected net profit. It will also connect farmers/FPOs with verified buyers, enable digital offers, lot creation, logistics coordination and payment tracking. A multilingual farmer-friendly interface, especially in Marathi, will make the platform easy to use and improve transparency, reduce transaction costs and increase farmers' price realization.
LRIH-13	HERTZ	SIH26229	Kabadiwala Connect – Bringing the Informal Collector into the Formal Recycling Chain	Scraplink informal e-waste collectors ko customer se connect karta hai
LRIH-14	CLUSTRA	SIH26131	Early detection and management of crop diseases and pest infestations	Crop-Guard AI is a software platform that combines crop images, weather, crop stage, field data and geo-tagged reports to detect diseases/pests and predict their future risk. It provides: AI-based disease & pest identification Weather-based risk forecasting GIS hotspot mapping Early alerts and localized IPM advisories Expert/lab referral for uncertain cases Agriculture officer dashboard for monitoring Feedback loop using field confirmations Core idea: Move from reactive disease detection to proactive early warning and targeted crop-health management.